

Task 4: Power BI Traffic Intelligence Dashboard

Import the traffic CSV file into **Power BI Desktop** and use **Power Query Editor** to prepare the data.

4.1 Data quality and preparation

Perform the following:

- Determine the number of rows and columns.
- Identify null/missing values and the columns affected.

- Check whether the data types are appropriate.
- Extract **Hour** from DateTime.
- Create **Temperature in Celsius**.
- Create a **Traffic Category**:

Traffic Volume	Category
< 4,500	Low
4,500–5,500	Medium
> 5,500	High

Requirement	Result
Original dataset size	48,204 rows × 9 columns
Null/missing values	No empty/null values identified
Data quality	100% valid, no Power Query null/type errors, 0% empty
Data types	Reviewed and set appropriately
date_time	Date/Time
traffic_volume	Whole Number
temp	Decimal Number
Hour	Created from date_time, Whole Number
Temperature_C	Created as temp – 273.15, Decimal Number
Traffic Category	Created as Low / Medium / High, Text

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Transpose Reverse Rows

Data Type: Text Replace Values Unpivot Columns

Detect Data Type Fill Move

Rename Pivot Column Convert to List

Split Column Format

Merge Columns Extract Parse

Statistics Standard Scientific Rounding Information

Date Time Duration

Run R Run Python script

Queries [1]

Metro_Interstate_Traffic_...

Table.TransformColumnTypes(#"Added Conditional Column",{"Traffic Category", type text})

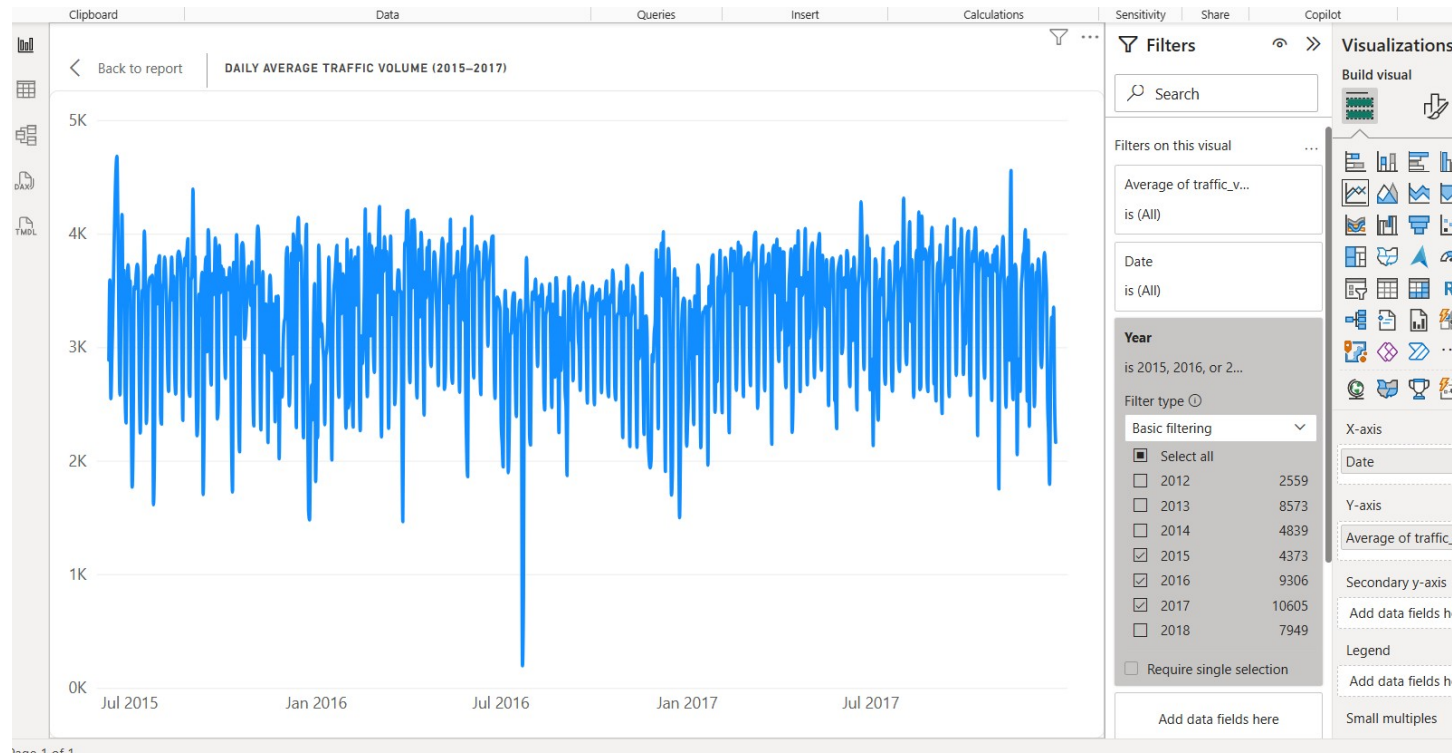
	weather_description	date_time	traffic_volume	Hour	Temperature_C	Traffic Category
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty
	38 distinct, 1 unique	40575 distinct, 35130 unique	6704 distinct, 432 unique	24 distinct, 0 unique	5843 distinct, 648 unique	3 distinct, 0 unique
1	scattered clouds	2/10/2012 9:00:00 am	5545	9	15.13	High
2	broken clouds	2/10/2012 10:00:00 am	4516	10	16.21	Medium
3	overcast clouds	2/10/2012 11:00:00 am	4767	11	16.43	Medium
4	overcast clouds	2/10/2012 12:00:00 pm	5026	12	16.98	Medium
5	broken clouds	2/10/2012 1:00:00 pm	4918	13	17.99	Medium
6	sky is clear	2/10/2012 2:00:00 pm	5181	14	18.57	Medium
7	sky is clear	2/10/2012 3:00:00 pm	5584	15	20.02	High
8	sky is clear	2/10/2012 4:00:00 pm	6015	16	20.71	High
9	few clouds	2/10/2012 5:00:00 pm	5791	17	20.99	High
10	few clouds	2/10/2012 6:00:00 pm	4770	18	19.95	Medium
11	few clouds	2/10/2012 7:00:00 pm	3539	19	17.82	Low
12	sky is clear	2/10/2012 8:00:00 pm	2784	20	16.23	Low
13	sky is clear	2/10/2012 9:00:00 pm	2361	21	15.46	Low
14	sky is clear	2/10/2012 10:00:00 pm	1529	22	14.01	Low
15	sky is clear	2/10/2012 11:00:00 pm	963	23	12.3	Low
16	sky is clear	3/10/2012 12:00:00 am	506	0	11.48	Low
17	sky is clear	3/10/2012 1:00:00 am	321	1	10.32	Low
18	sky is clear	3/10/2012 2:00:00 am	273	2	8.03	Low
19	sky is clear	3/10/2012 3:00:00 am	367	3	7.94	Low
20	sky is clear	3/10/2012 4:00:00 am	814	4	6.38	Low
21	sky is clear	3/10/2012 5:00:00 am	2718	5	5.47	Low

4.2 Dashboard analysis

Create visualisations showing:

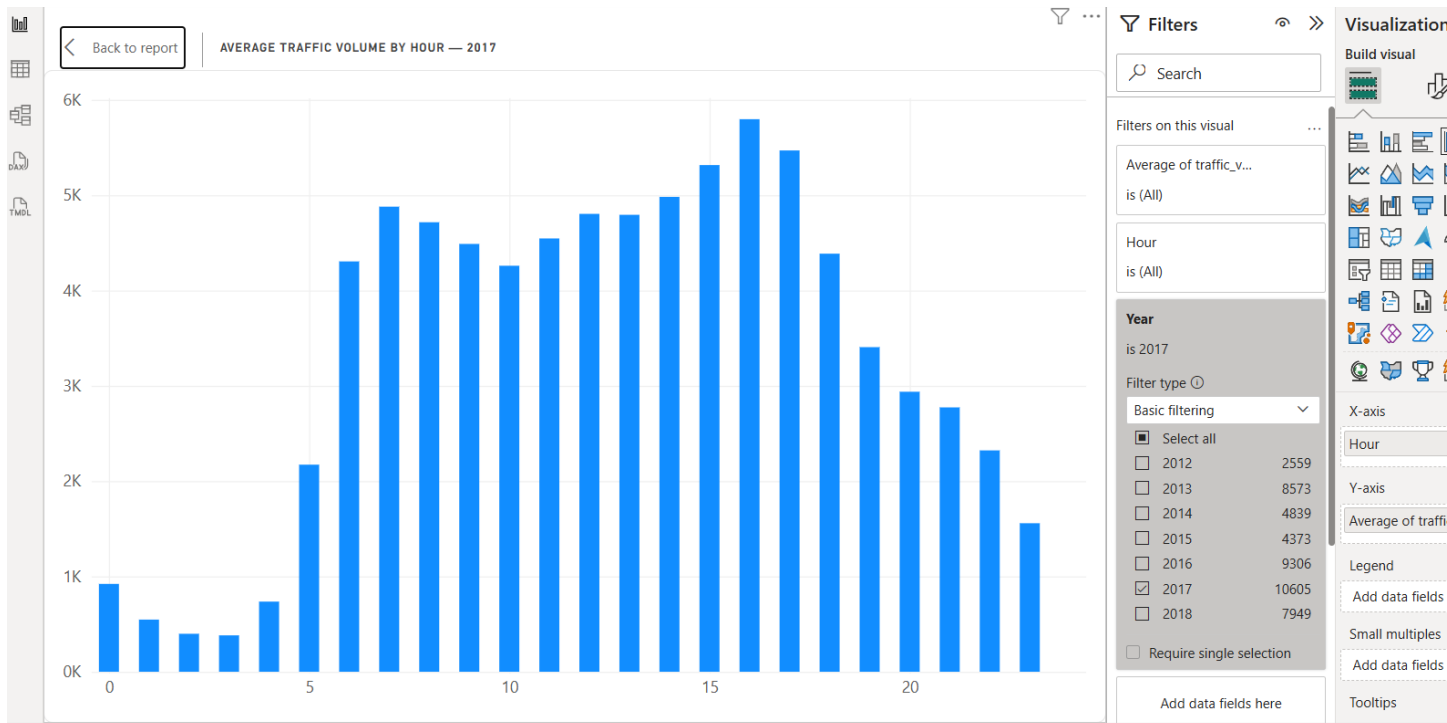
A. Daily traffic trends

- Show daily average traffic trends for 2015, 2016 and 2017.



B. Hourly traffic patterns

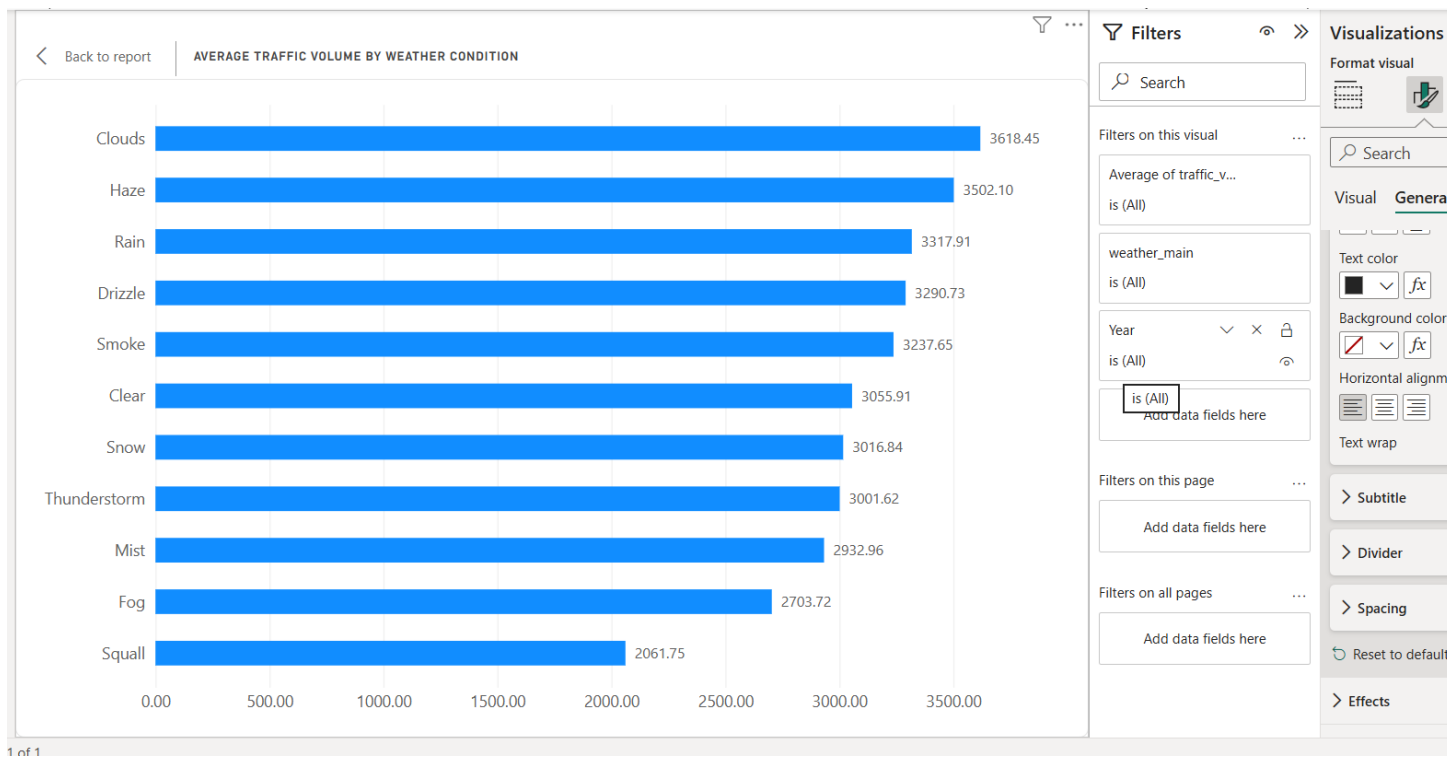
- Create a column chart showing average traffic by hour for 2017.



C. Weather impact

Analyse average traffic by weather condition and answer:

- Which weather condition has the highest average traffic? - Clouds
- Which has the lowest? - Squall
- What is the difference between the highest and lowest average traffic? $3,618.45 - 2,061.75 = 1,556.70$ vehicles



D. Temperature and traffic

Create a scatter plot of:

- Temperature vs. traffic volume.

Discuss:

- Whether a visible relationship exists.

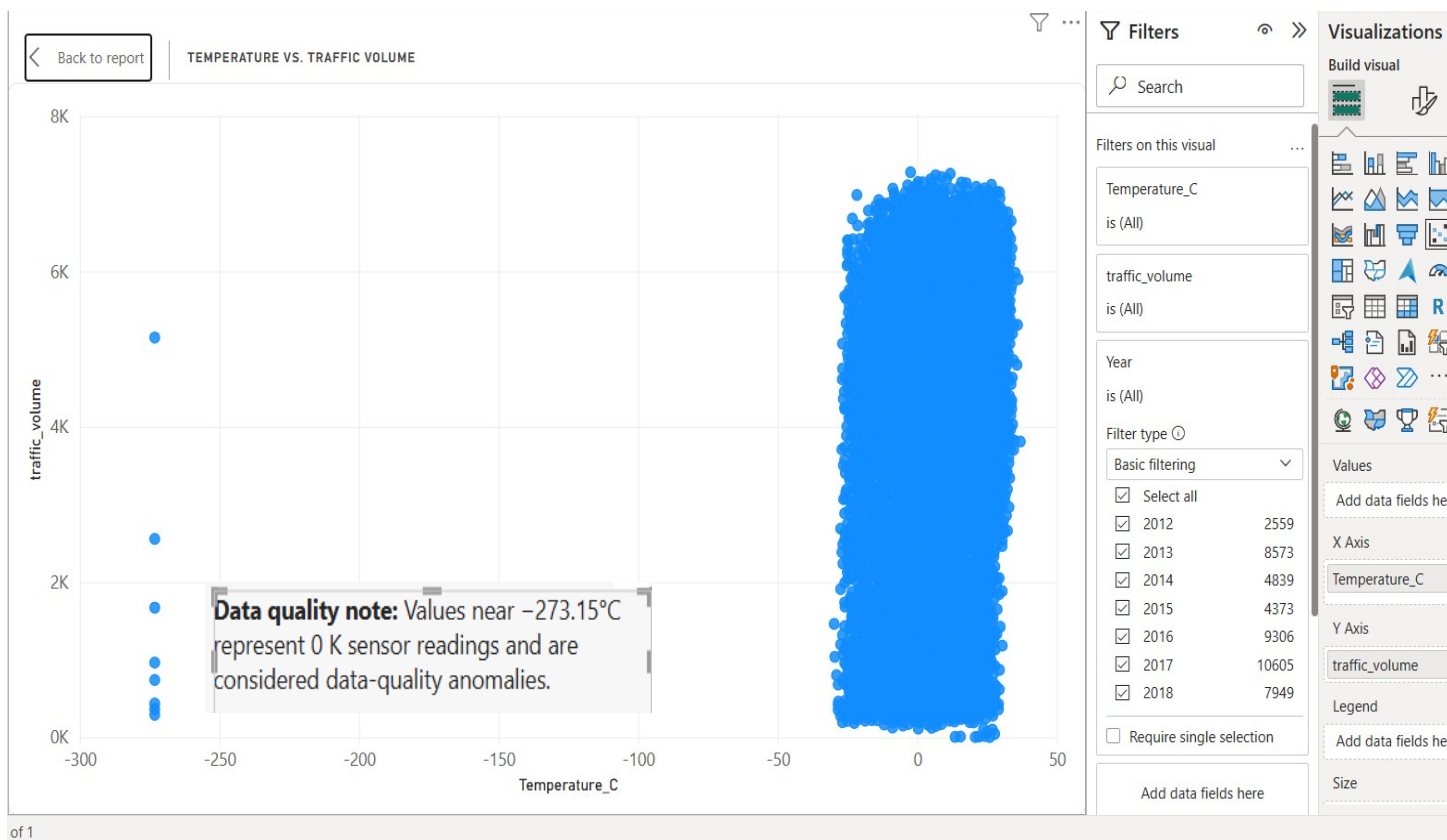
The scatter plot does not show a strong linear relationship between temperature and traffic volume. Although the Pearson correlation coefficient of **0.1303** indicates a weak positive association, traffic volume varies considerably at similar temperatures. This suggests that temperature alone has limited ability to explain traffic demand.

- The temperature range associated with higher traffic.

Higher traffic volumes are mainly observed within the normal temperature range of approximately **-30°C to +35°C**. However, high traffic occurs across many temperatures within this range, so there is no clear temperature band that consistently produces high traffic.

- Any notable outliers.

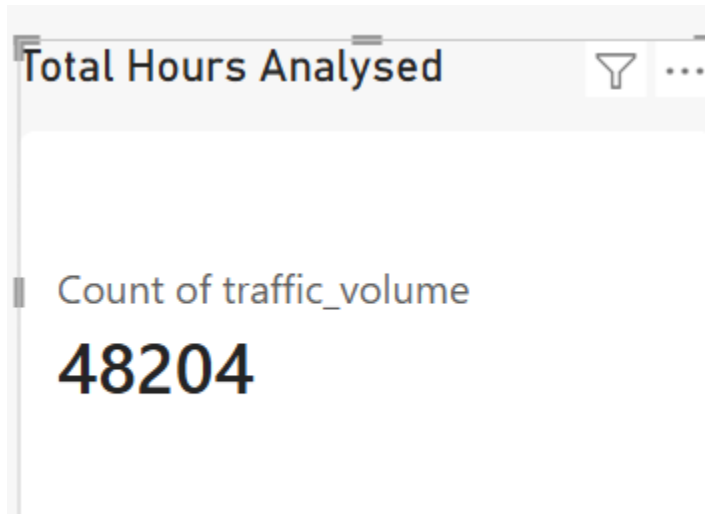
A notable group of outliers appears at approximately **-273.15°C**, corresponding to temperature readings of **0 Kelvin**. These values are physically implausible for ambient weather and likely represent erroneous or missing sensor readings. They should therefore be treated as data-quality anomalies rather than genuine temperatures. These outliers are retained here to highlight the issue and should be addressed during subsequent data cleaning.



4.3 KPI cards and filters

Create KPI cards showing:

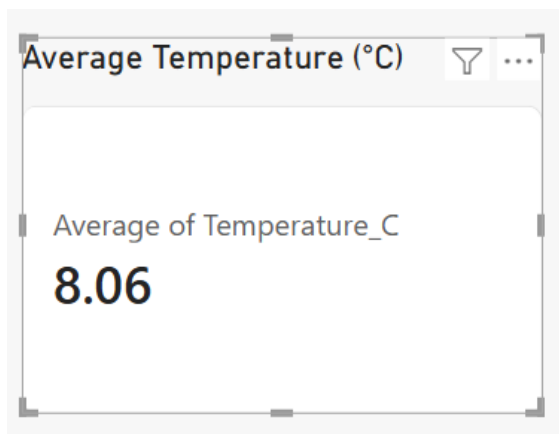
- Total hours analysed



- Average traffic volume



- Average temperature



Add slicers for:

- Hour range

The image shows a 'Hour Range' slicer interface. At the top, the title 'Hour Range' is displayed next to a set of icons: a funnel, a diamond, a square with an arrow, and a three-dot menu. Below the title, the word 'Hour' is centered. Underneath, there are two input boxes; the first contains the number '0' and the second contains '23'. Below these boxes is a horizontal range slider with two circular handles at the ends, connected by a solid black line. The entire interface is set against a light gray background.

- Weather condition

Weather Condition

weather_main

☐ Clear

☐ Clouds

☐ Drizzle

☐ Fog

☒ Haze

☐ Mist

☐ Rain

☐ Smoke

- Traffic category

Traffic Category

Traffic Category

☒ High

☐ Low

☐ Medium